What window telemetry can and cannot reveal about quantum hacking: a finite-key, drift-aware benchmark of intrusion detection for decoy-state QKD

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Abstract

Machine-learning intrusion detectors for quantum key distribution (QKD) report near-perfect accuracy on simulated attacks, yet their benign data are mostly stationary and their attackers are not tuned to the monitored statistics. We ask what window-level telemetry of a three-intensity decoy-state BB84 link can reveal. A faked-state attacker who reproduces the honest photon-number yields and error probabilities leaves the joint law of all detection and error counts unchanged; every telemetry detector then operates at chance while the finite-key analysis outputs a key the attacker knows. For attacks inside the device model the decoy-state bound already protects the key. We propose a decoy-consistency statistic (DCS) that refits the drifting channel parameters in every window and tests only the relations that drift cannot change, and we evaluate it with five baselines on 24-hour streams under four drift scales with an exact finite-key audit of attacker knowledge. Over 10 seeds, isolation forests and random forests calibrated on stationary data alarm in 97.15% and 97.46% of benign windows under nominal drift; domain-randomised training lowers this to 9.16%. The DCS stays between 0.08% and 0.18% up to fourfold drift and flags naive and gain-only blinding in the first 10-second window. All detectors, the DCS included, score an AUC between 0.47 and 0.55 on the calibrated attack, and every key block produced under any blinding variant was fully known to the attacker. Telemetry monitoring can catch imperfectly calibrated attacks without drowning operators in false alarms, but it cannot replace hardware countermeasures against detector control.

Keywords: quantum key distribution, decoy states, detector blinding, intrusion detection, concept drift, finite-key security

1 Introduction

Quantum key distribution (QKD) moves the security argument from computational hardness to physics. Its information-theoretic guarantee holds only inside a device model; once the hardware departs from that model, an eavesdropper can learn the key while every monitored statistic looks normal [1, 2]. Detector blinding is the standard example. Bright illumination turns single-photon avalanche detectors into classical threshold detectors, and a faked-state eavesdropper

then dictates which detector clicks [3, 4]; a full-field implementation against a running link left the quantum bit error rate (QBER) unchanged and gave the attacker the entire key [5].

Operators therefore watch telemetry. A deployed decoy-state BB84 link [6–9] reports, per monitoring window, detection counts and error counts for every intensity and basis, and long field trials already feed some of these signals back into polarisation and phase stabilisation [10–12]. A recent strand of work trains machine-learning (ML) classifiers on such telemetry and reports near-perfect detection of intercept-resend, photon-number-splitting (PNS), Trojan-horse and blinding attacks [13–17]. These results come from simulators. In most of them the benign class is a stationary distribution, the classes are balanced, and the attacker is not calibrated against the statistics the defender monitors.

Two questions follow and neither has a quantitative answer in that literature. The first concerns the attacker: which attacks are visible in window telemetry at all, given that a capable eavesdropper can shape the statistics she produces? The second concerns the defender: how many false alarms does a telemetry detector raise when the fibre drifts for a day, as installed fibre does [10, 11], rather than sitting at its calibration point? A detector that alarms every few minutes on a healthy link will be switched off; a detector that misses the one attack that breaks the security proof adds nothing that the decoy-state parameter estimation does not already provide.

This paper answers both questions for three-intensity decoy-state BB84 with the finite-key analysis of Lim et al.[18]. The contributions are four.

- An indistinguishability result (Proposition 1). A faked-state attacker with a photon-number-resolving detector and knowledge of the channel can reproduce the honest photon-number yields and error probabilities exactly; the joint law of all window telemetry is then identical to the honest law, every telemetry detector operates on the chance diagonal of its ROC curve, and the finite-key analysis outputs a key that the attacker knows completely.

- A decoy-consistency statistic (DCS) that refits the honest channel model in every window and tests the remaining degrees of freedom. Slow drift of the channel parameters is absorbed by the refit. The statistic needs only benign drift-free calibration data and costs $O(1)$ per window.

- An open, finite-key-aware benchmark. It covers ten attacker configurations under four drift scales over 24-hour streams; a ladder of partially calibrated blinding attacks is part of the set. It reports per-window false alarms, ROC AUC and, for every post-processing block, the finite-key length together with an exact count of key bits known to the attacker.

- A negative result for the ML baselines. Classifiers and isolation forests calibrated on stationary data raise alarms in almost every benign window once realistic drift is present; drift-randomised training reduces but does not remove the effect, and none of the detectors, ours included, separates the calibrated blinding attack from honest operation.

The scope is deliberately narrow. All data are simulated, the device model is the standard one used in decoy-state analyses [18, 19], and the claims are about what window-level telemetry can certify. Device-level countermeasures, such as measurement-device-independent QKD [20]

or detector monitoring [21, 22], are outside the detectors studied here and are discussed in Section 8.

Section 2 reviews decoy-state QKD together with prior work on quantum hacking and its detection. Section 3 fixes the channel and threat models. Section 4 states what telemetry can and cannot reveal. Section 5 defines the DCS. Sections 6 and 7 give the experimental design and the results, and Section 8 discusses limitations.

2 Background and related work

2.1 Decoy-state BB84 and finite-key security

Weak coherent pulses carry a Poisson photon number, so multi-photon pulses leak information through photon-number splitting [23]. The decoy-state method [7–9] sends several intensities at random; since an eavesdropper sees only the photon number, the yields y_n and error rates e_n cannot depend on the intensity, and linear programming over the observed gains bounds the single-photon yield and its phase error. Practical versions use one signal, one weak decoy and one vacuum-like intensity [19]. Lim et al.[18] give closed-form finite-key bounds for this three-intensity protocol that hold against general attacks, using Hoeffding-type deviations and a sampling bound for the phase error; tighter analyses exist [24], but the closed form suits a benchmark that evaluates thousands of blocks.

2.2 Quantum hacking

Implementation attacks exploit the gap between device and model [1, 2, 25]. Time-shift attacks use detector efficiency mismatch [26]; Trojan-horse attacks read the modulator settings through back-reflected light [27, 28]; blinding attacks remove the quantum behaviour of the detectors altogether [3–5]. Countermeasures act on the hardware: photocurrent monitoring [21], random detector efficiency [22], whose assumptions were later broken in a commercial system [29], and measurement-device-independent QKD [20], which removes detector side channels by design.

2.3 Telemetry-based attack detection

Machine-learning detectors were first proposed for continuous-variable QKD, where the shot-noise and excess-noise estimates form natural features [13, 14]. For discrete-variable QKD, Al-Kuwari et al.[15] train hybrid quantum-classical LSTMs on 10 329 simulated decoy-state samples over eight balanced scenarios; Isha et al.[16] classify 24 000 simulated BB84 sessions with gradient-boosted trees and report a false-negative rate of 0.0198 against 0.8477 for an 11% QBER threshold; Mohamed and Al-Kuwari [17] add domain randomisation of transmittance and misalignment and train against an optimising adversary. Machine learning has also been used on the defender's side for parameter optimisation [30]. Three gaps recur. The benign data are either stationary or randomised per sample, so the false-alarm rate under slow, correlated drift over a day is not measured; the attackers are not tuned to reproduce the monitored statistics; and detection is not tied to the finite-key output, so a reader cannot tell whether a detected attack would have compromised any key.

2.4 Drift and false alarms

Field links drift. Polarisation and fibre length change over tens of minutes, and long-running systems close feedback loops on the QBER and the count rates to compensate [10]; network trials have since run QKD layers in field and production settings, the SwissQuantum network for close to two years [12, 31], so a monitor must cope with environmental variation over days rather than minutes. In the stream-learning literature this is concept drift [32]: a model fitted to one regime degrades when the input distribution moves. For a monitoring system the visible symptom is the false-alarm rate. Sequential change detection, from Page's CUSUM onwards [33], treats false alarms as a design constraint rather than a by-product; that view drives the design of the statistic in Section 5.

3 Channel and threat model

3.1 Protocol and telemetry

Alice emits phase-randomised weak coherent pulses at rate R with intensity $k \in \mathcal{K} = \{\mu_1, \mu_2, \mu_3\}$, $\mu_1 > \mu_2 + \mu_3$, $\mu_2 > \mu_3 \geq 0$, chosen with probabilities p_k, in basis Z (key) with probability p_Z or X (test) otherwise. Bob uses two threshold detectors. Time is divided into monitoring windows of length Δ. Window t produces the telemetry vector

$$\mathbf{T}_t = \{N_{b,k}(t),\, n_{b,k}(t),\, m_{b,k}(t)\}_{b\in\{Z,X\},\, k\in\mathcal{K}}, \tag{1}$$

the pulses sent together with the detections and bit errors per basis and intensity after sifting. A post-processing block aggregates B consecutive windows and runs the finite-key analysis of [18], which returns a key length ℓ (zero means abort).

3.2 Honest channel and benign drift

Conditioned on an n-photon pulse, the honest channel with transmittance η (fibre and detector), background yield Y_0 and misalignment e_d produces a click with probability and error probability

$$y_n = 1 - (1 - Y_0)(1 - \eta)^n, \qquad e_n y_n = e_0 Y_0 + e_d\,(y_n - Y_0), \tag{2}$$

with $e_0 = 1/2$ [19]. The gain of intensity k is $Q_k = \sum_n P(n|k)\, y_n = 1 - (1 - Y_0)e^{-\eta k}$. In window t the parameters drift as

$$L(t) = \alpha\ell_f + d\,[A_L \sin(\omega t + \phi) + U_L(t)] \ \text{dB}, \qquad \eta(t) = \eta_B\,[1 + d\,U_B(t)]\,10^{-L(t)/10}, \tag{3}$$

$$p_{dc}(t) = p_{dc}\,[1 + d\,A_D \sin(\omega t + \phi - 0.5)], \qquad e_d(t) = e_{\text{mis}} + d\,s_j\,(t - t_j), \quad t \in [t_j, t_j + T_c), \tag{4}$$

where $\omega = 2\pi/24\,\text{h}$, U_L and U_B are Ornstein–Uhlenbeck processes, and the misalignment grows linearly at a random slope s_j until an active compensation at period T_c resets it. The scale d switches drift off ($d = 0$, the calibration regime) or amplifies it. Independently of d, every window carries a relative intensity fluctuation $\mu_k \to \mu_k(1 + \xi_{k,t})$, $\xi_{k,t} \sim \mathcal{N}(0, \sigma_I^2)$, which the detectors do not observe. Table 1 lists the values. They are assumptions chosen to reproduce the qualitative

Table 1: Simulation parameters. Device values follow the numerical example of [18]; drift amplitudes are scaled by d.

Quantity	Symbol	Value
Pulse rate, window length	R, Δ	100 MHz, 10 s
Intensities, probabilities	μ_k, p_k	(0.5, 0.1, 2×10^{-4}), (0.7, 0.2, 0.1)
Key-basis probability	p_Z	0.9
Fibre length, attenuation	ℓ_f, α	50 km, 0.2 dB/km
Detector efficiency, dark count	η_B, p_{dc}	0.10, 6×10^{-7} per gate
Base misalignment	e_{mis}	1%
Diurnal loss amplitude; OU loss s.d., time constant	A_L; σ_L, τ_L	0.3 dB; 0.15 dB, 30 min
Detector-efficiency OU s.d. (relative), time constant	σ_B, τ_B	2%, 60 min
Dark-count diurnal swing (relative)	A_D	10%
Misalignment growth, compensation period	s_j, T_c	$U(0.5, 1.5)\times 1\%$/h, 1 h
Intensity noise (per window, relative)	σ_I	1%
Post-processing block	B	60 windows (6×10^{10} pulses)
Security parameters	$\varepsilon_{\mathrm{sec}}$, $\varepsilon_{\mathrm{cor}}$, f_{EC}	10^{-10}, 10^{-15}, 1.16
Stream length		8 640 windows (24 h)

behaviour reported for installed fibre, slow loss variation and misalignment growth between compensations [10, 11]; they are not fitted to a specific link, and Section 7.5 varies the most influential ones.

3.3 Threat model

Eve controls the quantum channel and may illuminate Bob's detectors. She cannot read Alice's intensity or basis choices before the public announcement, and she has no access to the classical post-processing beyond what is announced. Every attack is described by the photon-number functions (y_n, e_n, κ_n), where κ_n is the fraction of resulting clicks whose sifted bit Eve knows exactly. Table 2 lists the attacks. Where the cited attack descriptions match Bob's signal gain, so do ours.

4 What telemetry can and cannot reveal

Alice's photon-number distribution is fixed by her source, and each pulse is processed independently given its photon number. The window counts are therefore sums of independent Bernoulli variables whose parameters depend on the attack only through (y_n, e_n). This gives the following statement.

Proposition 1 (Telemetry indistinguishability)**.** *Let $(y_n(t), e_n(t))$ be the honest yields and error probabilities of window t, and let an attack induce $(\tilde{y}_n(t), \tilde{e}_n(t))$. If $\tilde{y}_n(t) = y_n(t)$ and $\tilde{e}_n(t) = e_n(t)$ for all n and t, the joint law of $(\mathbf{T}_1, \ldots, \mathbf{T}_W)$ under the attack equals the honest law. For every detector ϕ that maps telemetry to alarms, $\Pr_{\mathrm{attack}}[\phi = 1] = \Pr_{\mathrm{honest}}[\phi = 1]$; the ROC curve of ϕ is the diagonal, and the finite-key length ℓ has the same distribution in both cases.*

Proof. Given Alice's settings (b, k) for a pulse, its photon number is Poisson with mean k (with k the realised intensity) and independent of everything else. A click and its error indicator are then Bernoulli with parameters $\sum_n P(n|k)\tilde{y}_n$ and $\sum_n P(n|k)\tilde{y}_n\tilde{e}_n / \sum_n P(n|k)\tilde{y}_n$, independently

Table 2: Attacker configurations. "In model" marks attacks covered by the decoy-state security proof; the others violate its detector assumptions.

Name	Mechanism	In model
IR, f	Intercept-resend on a fraction f of non-empty pulses; resend intensity chosen so that Q_{μ_1} is unchanged	yes
PNS	Lossless channel; single photons blocked, one photon of each multi-photon pulse forwarded with the probability that matches Q_{μ_1} (capped at 1) [23]	yes
Loss $+\delta$	Extra attenuation of δ dB (denial of service)	yes
Blind-naive	Faked states [3, 4] from a threshold detector; forwarding probability matches Q_{μ_1}; no dark counts, no deliberate errors	no
Blind-gain	Faked states with a photon-number-resolving detector; reproduces every y_n but adds no errors	no
Blind-nodark	Reproduces the signal part of y_n and the misalignment errors; blinded detectors give no dark counts	no
Blind-cal	Reproduces (y_n, e_n) of the current honest channel exactly (Proposition 1)	no
Blind-cal-static	As Blind-cal, but for the nominal channel of Table 1 without drift	no

across pulses. These parameters coincide with the honest ones, so $(N_{b,k}, n_{b,k}, m_{b,k})$ have the same multinomial–binomial law in every window, and the same holds for any function of them, such as ϕ and ℓ. □

The statement is elementary, and the perfect eavesdropper of [5] is an instance of it; its use here is to bound what any telemetry detector, learned or not, can achieve. The hypothesis is achievable. A faked-state attacker who blinds Bob's detectors forces a click only when Bob's basis matches hers, which happens with probability 1/2. If she measures each pulse with a photon-number-resolving detector, forwards an n-photon outcome with probability $2y_n(t)$, injects a click for a fraction of empty pulses that reproduces Y_0, and flips the forwarded bit with probability $e_n(t)$, she meets the hypothesis whenever $2y_n(t) \leq 1$; for the parameters used here this holds for every n whose Poisson weight exceeds 10^{-30}. After sifting, Alice's basis equals Bob's and hence Eve's, so Eve knows every sifted bit; she also knows which of them she flipped. The attack requires knowledge of $\eta(t)$, $Y_0(t)$ and $e_d(t)$; an eavesdropper who owns the fibre can estimate them from her own measurements of the line, and Section 7 also evaluates an attacker who ignores drift.

Remark 1 (In-model attacks)**.** *Intercept-resend and PNS act only on the quantum channel, as does added loss. The finite-key bound of [18] is $\varepsilon_{\mathrm{sec}}$-secret against such attacks, so every key it outputs is already secure; a telemetry detector can shorten the time to incident response or flag a denial of service, but it cannot add secrecy. Conversely, for attacks outside the model the bound can certify a key that Eve knows. Telemetry detection matters for security only in the second case, and Proposition 1 limits it to attackers who do not reproduce the honest statistics.*

To audit secrecy in the simulations without trusting the bound, every click carries a ground-truth flag stating whether Eve knows its sifted bit. With K_E known bits among the n_Z sifted key bits of a block, the smooth min-entropy of the sifted key given Eve's information is at most

$n_Z - K_E$ up to terms of order $\log(1/\varepsilon)$, so a block with

$$\ell > n_Z - K_E \tag{5}$$

cannot be ε-secret for small ε. We call such a block a *breach*. The test is one-sided: $\ell \le n_Z - K_E$ does not prove secrecy, because Eve may hold further partial information. It also checks the bound itself, since the true single-photon detections are recorded and must exceed the lower bound $s_{Z,1}$ in honest operation.

5 Decoy-consistency statistic

Drift moves three physical parameters; the telemetry of one window has twelve count pairs. The DCS spends three degrees of freedom on tracking the channel and tests the rest. Let $Q_k = \sum_b n_{b,k} / \sum_b N_{b,k}$ be the pooled gain of intensity k. A fixed-point iteration on the signal and vacuum-like gains,

$$\hat{\eta} = -\frac{1}{\mu_1} \ln \frac{1 - Q_{\mu_1}}{1 - \hat{Y}_0}, \qquad \hat{Y}_0 = \frac{Q_{\mu_3} - (1 - e^{-\hat{\eta}\mu_3})}{e^{-\hat{\eta}\mu_3}}, \tag{6}$$

started from $\hat{Y}_0 = Q_{\mu_3}$ and run for four steps, gives $(\hat{\eta}, \hat{Y}_0)$; the signal error count then gives $\hat{e}_d = (m_{\mu_1}/N_{\mu_1} - e_0\hat{Y}_0)/(Q_{\mu_1} - \hat{Y}_0)$, clipped to $[0, 1/2]$. With these estimates the honest model (2) predicts the remaining observations, and six binomial residuals

$$r_j = \frac{c_j - M_j \hat{p}_j}{\sqrt{M_j \hat{p}_j (1 - \hat{p}_j)}}, \qquad j = 1, \ldots, 6, \tag{7}$$

compare an observed count c_j out of M_j trials with its predicted probability $\hat{p}_j$. The six are the pooled decoy gain, the decoy error rate in each basis, the vacuum-like error rate, the signal error rate in the test basis, and the share of signal detections in the key basis (which must equal the share of pulses sent in that basis). The decoy-state estimator uses the same relations to bound the single-photon contribution over a whole block; the DCS turns them into a per-window test with a controlled false-alarm rate. The statistic is

$$S_t = \sum_{j=1}^{6} \left(\frac{r_j - \mathrm{med}_j}{1.4826\,\mathrm{MAD}_j} \right)^2, \tag{8}$$

where the median and median absolute deviation come from benign windows recorded without slow drift. The robust standardisation absorbs the fact that unobserved intensity noise inflates the decoy-gain residual beyond its binomial variance. An alarm is raised when S_t exceeds the $(1 - \alpha)$ empirical quantile of S on separate benign calibration windows.

Each window costs a constant number of arithmetic operations, and the calibration needs only benign data at the operating point. Slow drift in η, Y_0 or e_d moves the estimates, not the residuals. Two blind spots are built in. Pure added loss is indistinguishable from a transmittance change, so the DCS cannot flag it; the key rate shows such an attack anyway. An intercept-resend attack on a small fraction raises the signal QBER and is partly absorbed into $\hat{e}_d$, so only

the decoy and test-basis residuals see it.

6 Experimental design

6.1 Streams and protocol

Each realisation (seed) generates its own channel trajectory, calibration data, training data and test streams; seeds are the unit of replication, and all intervals below are 95% t-intervals over 10 seeds. A test stream covers 24 hours (8 640 windows). Benign streams are run at drift scales $d \in \{0, 1, 2, 4\}$; for each scale and each attack of Table 2, a further stream carries the attack for two hours (720 windows) starting at a uniformly random window between hour 4 and hour 20. Counts are drawn per photon-number component ($n = 0$, $n = 1$, $n \geq 2$), so every block records its true single-photon detections and the number of sifted key bits Eve knows. The finite-key analysis runs on every complete block of 60 windows that lies inside the attack, and on all 144 blocks of each benign stream.

6.2 Detectors

Six detectors share one interface: a score per window and a threshold.

- **QBER>11%**: the key-basis QBER of the signal intensity against the fixed 11% limit used as the baseline in [16].
- **DCS**: Section 5, standardised on one drift-free benign stream.
- **IF-stat** and **IF-drift**: isolation forests [34] (200 trees) on nine features, the logarithm of the pooled gain per intensity and the QBER per basis and intensity. IF-stat is fitted on one drift-free benign stream; IF-drift on three benign streams with $d \in \{0.5, 1, 2\}$, which gives it representative drift that the other unsupervised detectors do not see.
- **RF-stat** and **RF-DR**: random forests [35] (200 trees, minimum leaf size 2) on the same features, trained as binary attack-versus-benign classifiers in the style of the literature [15, 16]. The training data hold one benign stream and nine attack streams of 2 160 windows each (IR with $f \in \{0.05, 0.1, 0.25, 0.5\}$, PNS, Blind-naive, loss of 1, 3 and 6 dB). RF-stat sees no drift; RF-DR draws a drift scale from $U(0, 2)$ for every training stream, the domain randomisation of [17]. Neither sees Blind-gain, Blind-nodark or the calibrated attacks.

Every threshold except the fixed QBER limit is the $(1 - \alpha)$ quantile, $\alpha = 10^{-3}$, of the detector's score on three further drift-free benign streams (25 920 windows). This is the default an operator would deploy after commissioning. As a control, thresholds are also set on three benign streams with drift ($d \in \{0.5, 1, 2\}$); this uses benign drift data that a new link does not have, and is reported separately. All settings were fixed before the test streams were generated; the only change after the first full run was the post-hoc DCS+Y0 variant of Section 7.5, which is evaluated on fresh seeds.

6.3 Metrics and statistics

For each detector and stream the benchmark records the per-window false-alarm rate on benign streams (FPR), the true-positive rate on attacked windows at the calibrated threshold (TPR), the ROC AUC of attacked against non-attacked windows of the same stream, and the delay from attack onset to the first alarm. For each block it records ℓ, n_Z, K_E, the breach indicator (5) and the bound check $s_{Z,1} \leq$ true single-photon count. Paired comparisons between DCS and each other detector use two-sided Wilcoxon signed-rank tests [36] over seeds with Holm correction [37] across the five comparisons. With ten seeds the smallest attainable two-sided p is $2^{-9} \approx 0.0020$, and 0.0098 after Holm correction; p-values at that floor indicate that DCS was better in every seed, not the size of the difference, which the intervals report. One seed of the full benchmark takes about 20 s on a single CPU core with four threads for the forests.

7 Results

7.1 Honest operation and the bound check

The simulator reproduces the expected finite-key behaviour. At the calibration point the honest link yields 1.41×10^{-3} secret bits per pulse; nominal drift lowers this to 1.21×10^{-3} and fourfold drift to 0.76×10^{-3}, where 95.7% of the blocks still produce a key. Over the 5760 honest blocks the lower bound $s_{Z,1}$ exceeded the true single-photon count 0 times, and no honest block met the breach condition (5). At $d = 1$ the stream loss spans 9.2–10.8 dB and the misalignment reaches 2.5%; at $d = 4$ the ranges are 7.1–12.8 dB and 7.0%.

7.2 False alarms under drift

Table 3 and Figure 1 give the per-window false-alarm rate on benign streams. Every calibrated detector meets its 10^{-3} target at $d = 0$. Drift separates them at once. IF-stat and RF-stat alarm in 97.15% and 97.46% of benign windows at $d = 1$; their alarms track the hourly misalignment cycle (Figure 2, lower panels), so an operator would see a periodic alarm on a healthy link. Domain randomisation helps but does not hold the target: RF-DR reaches 9.16% at $d = 1$ and 64.82% at $d = 4$, with large seed-to-seed spread. IF-drift, fitted on benign drift up to $d = 2$, stays at 0.34% at $d = 1$ and rises to 29.82% and 78.87% at $d = 2$ and $d = 4$. The DCS stays between 0.08% and 0.18% at every scale. Its paired difference from each learned detector at $d \geq 1$ has Holm-adjusted $p \leq 0.0098$; against the fixed QBER rule, which raised no alarm at all, the DCS is the worse of the two on this metric. Moving the thresholds to drifted calibration data (Table 7 in the appendix) brings RF-DR to 5.88% and IF-drift to 22.17% at $d = 4$, at the price of benign drift data recorded on the target link.

7.3 Which attacks are separable

Figure 3 gives the ROC AUC at $d = 1$; Table 8 in the appendix gives $d = 0$. Four results stand out.

The calibrated attack is invisible. For Blind-cal the AUC of every detector at every drift scale lies between 0.47 and 0.55, as Proposition 1 requires. The high TPR of IF-stat and RF-stat

Table 3: Per-window false-alarm rate (%) on 24-hour benign streams, thresholds calibrated on drift-free data for $\alpha = 0.1\%$. Mean ± 95% t-interval over 10 seeds.

Detector	$d = 0$	$d = 1$	$d = 2$	$d = 4$
QBER>11%	0.00	0.00	0.00	0.00
DCS (ours)	0.09±0.02	0.08±0.04	0.11±0.03	0.18±0.06
IF-stat	0.13±0.03	97.15±0.20	99.09±0.11	99.78±0.05
IF-drift	0.11±0.04	0.34±0.19	29.82±3.63	78.87±2.90
RF-stat	0.12±0.03	97.46±0.26	99.15±0.11	99.72±0.06
RF-DR	0.11±0.03	9.16±12.36	33.65±19.94	64.82±14.67

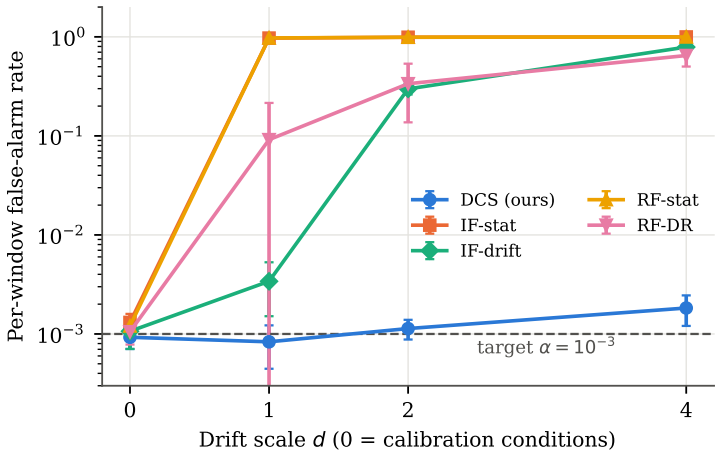


Figure 1: Per-window false-alarm rate against drift scale (log axis; error bars are 95% t-intervals over 10 seeds). IF-stat (squares) and RF-stat (triangles) overlap near 1 for $d \geq 1$. The QBER rule raised no alarms and is not shown on the log axis. The dashed line is the calibration target.

on Blind-cal in Table 4 is their benign false-alarm rate, not detection.

Errors in the attacker's calibration are visible to the DCS when they break a physical relation the refit cannot absorb. Blind-naive (no dark counts, no errors, wrong decoy ratio) and Blind-gain (correct gains, no errors) give AUC 1.00 and an alarm in the first attacked window in every seed and at every drift scale. Blind-nodark is the exception: the refit lowers $\hat{Y}_0$ to match the missing dark counts and the residuals stay small, so the DCS scores 0.55 at $d = 1$. The learned detectors see this attack (IF-drift 1.00, RF-DR 0.93) because they compare absolute levels with what they were trained on.

An attacker who ignores drift becomes distinguishable in the opposite direction. Blind-cal-static emits the statistics of the nominal channel while the real channel drifts. The learned detectors score these windows as *more* normal than the benign ones (IF-stat AUC 0.00, RF-stat 0.00 at $d = 1$), so the drift itself is the signature, but only a detector that alarms on missing variation would use it. The DCS, which refits every window, scores 0.50.

In-model attacks are detected but do not need to be. IR and PNS give DCS AUC near 1 at every drift scale; at $d = 0$ the learned detectors reach AUC between 0.91 and 1.00 on the same attacks. The DCS cannot see added loss (AUC below 0.5) by construction.

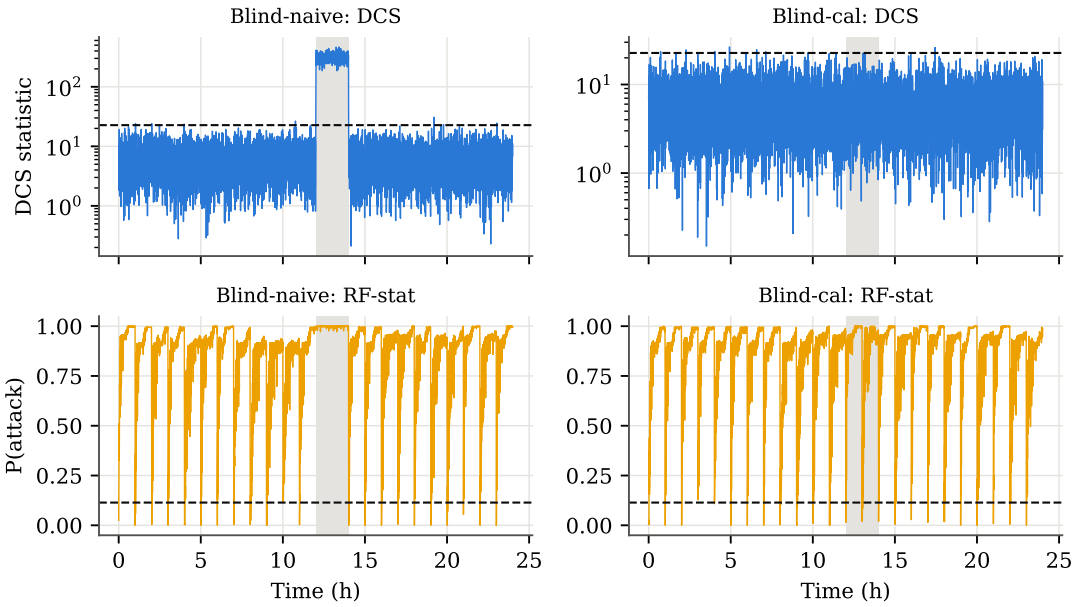


Figure 2: Window scores over one 24-hour stream at $d = 1$ with the attack active in the shaded two hours; dashed lines are the calibrated thresholds. Left: Blind-naive. Right: Blind-cal. Top: DCS (log axis). Bottom: RF-stat, whose score follows the hourly misalignment ramp and crosses its threshold in most benign windows.

7.4 Finite-key audit

Table 5 links detection to secrecy at $d = 1$. The finite-key analysis aborts every PNS block and every IR block at $f = 0.2$; at $f = 0.05$ it outputs a shorter key (0.80×10^{-3} per pulse) while Eve knows 2.5% of the sifted bits, and no block breaches. Added loss only shortens the key. No in-model attack produced a breach in any block at any drift scale (maximum 0.0%). Every blinding variant did the opposite: in every block where the analysis produced a key, that key was fully known to Eve (at least 94.2% of attacked blocks at every scale; the remainder are blocks that aborted because of drift). Blind-naive even raises the key rate to 2.56×10^{-3}, since its faked clicks carry no errors. The DCS flags Blind-naive and Blind-gain within the first 10-second window, long before the first 10-minute block closes, so those blocks can be discarded; it does not flag Blind-nodark, Blind-cal or Blind-cal-static.

7.5 Sensitivity and a post-hoc variant

Table 6 varies one factor at a time on separate seeds (five per setting, $d = 1$, 8-hour streams). The false-alarm rate stays between 0.04% and 0.17% throughout. Detection power depends on counts per window. With 1-second windows the DCS still flags Blind-naive but loses Blind-gain (TPR 0.00) and IR at $f = 0.05$ (TPR 0.00); with 60-second windows the IR TPR rises to 1.00 and Blind-gain stays at 1.00. At 75 km the IR TPR falls to 0.21. Intensity noise between 0.5% and 2% has little effect, since the robust standardisation absorbs it.

The Blind-nodark result suggested one change: add the background-yield estimate $\hat{Y}_0$ itself as a seventh standardised residual, since blinded detectors produce no dark counts and the fitted level should not leave its calibrated range. This DCS+Y0 variant was defined after the main run

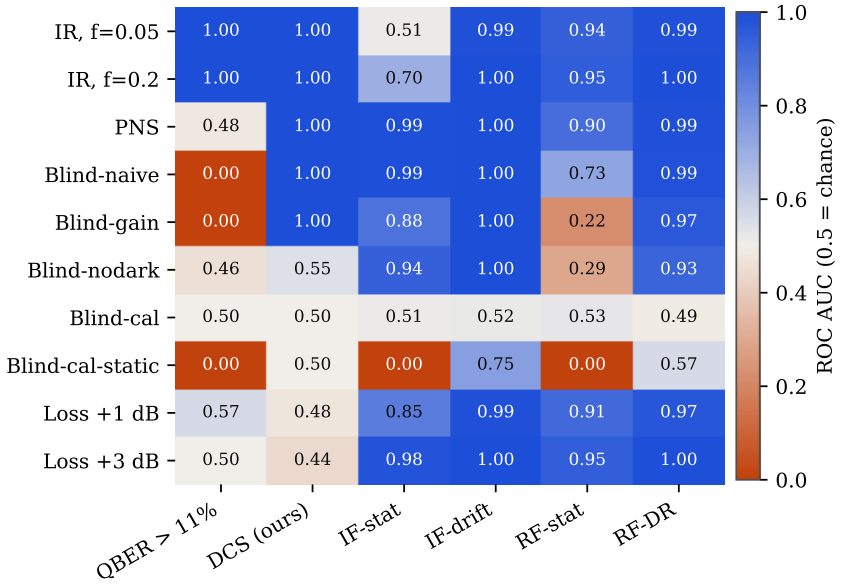


Figure 3: ROC AUC at $d = 1$ (mean over 10 seeds). Blue: attack windows score higher than benign windows; orange: lower; light grey: chance. Values below 0.5 for the static calibrated attack mean that its windows look more stationary than the drifting benign windows.

Table 4: TPR at the drift-free calibrated threshold, $d = 1$ (mean over 10 seeds). The first row is the benign FPR of the same detector; a TPR close to it is not detection.

Attack	QBER>11%	DCS (ours)	IF-stat	IF-drift	RF-stat	RF-DR
benign FPR	0.000	0.001	0.972	0.003	0.975	0.092
IR, $f = 0.05$	0.000	0.946	1.000	0.560	1.000	0.883
IR, $f = 0.2$	0.000	1.000	1.000	1.000	1.000	1.000
PNS	0.000	1.000	1.000	1.000	1.000	1.000
Blind-naive	0.000	1.000	1.000	1.000	1.000	0.997
Blind-gain	0.000	1.000	1.000	0.997	1.000	0.969
Blind-nodark	0.000	0.002	1.000	0.940	1.000	0.846
Blind-cal	0.000	0.001	0.974	0.003	0.976	0.054
Blind-cal-static	0.000	0.001	0.001	0.001	0.000	0.001
Loss +1 dB	0.000	0.000	0.997	0.707	1.000	0.875
Loss +3 dB	0.000	0.001	1.000	1.000	1.000	1.000

and was evaluated on ten fresh seeds. It lifts the Blind-nodark AUC from 0.54 to 1.00 at $d = 1$ and keeps the false-alarm rate at 0.11% there, but the rate rises to 1.82% at $d = 4$ against 0.20% for the DCS, because the dark-count swing scales with d. Neither variant separates Blind-cal (AUC near 0.5).

8 Discussion

8.1 What a telemetry detector is for

The finite-key audit splits the attacks cleanly. Attacks inside the device model never produced a breach; the decoy-state bound shortened or aborted the key on its own, as Remark 1 predicts. Attacks outside the model breached every block that produced a key. A telemetry detector therefore adds secrecy only against model-violating attacks, and Proposition 1 limits even that

Table 5: Finite-key audit at $d = 1$ over all complete blocks inside the attack (120 blocks per scenario, 1 440 for the honest stream). "Breach" is the share of blocks satisfying (5).

Scenario	$P(\ell > 0)$	Rate ($\times 10^{-3}$)	Eve-known fraction	Breach
Honest (24 h)	1.000	1.214	0.000	0.000
IR, $f = 0.05$	1.000	0.797	0.025	0.000
IR, $f = 0.2$	0.000	0.000	0.101	0.000
PNS	0.000	0.000	1.000	0.000
Blind-naive	1.000	2.559	1.000	1.000
Blind-gain	1.000	1.985	1.000	1.000
Blind-nodark	1.000	1.199	1.000	1.000
Blind-cal	1.000	1.195	1.000	1.000
Blind-cal-static	1.000	1.408	1.000	1.000
Loss +1 dB	1.000	0.929	0.000	0.000
Loss +3 dB	1.000	0.584	0.000	0.000

Table 6: One-factor sensitivity of the DCS at $d = 1$ (five seeds per row; base setting: 10 s, $\sigma_I = 1\%$, 50 km). Columns 3–6 are TPR at the calibrated threshold.

Setting	FPR (%)	IR f=0.05	Blind-naive	Blind-gain	Blind-nodark
Window 1 s	0.04	0.000	0.989	0.000	0.000
Window 10 s	0.10	0.932	1.000	1.000	0.002
Window 60 s	0.17	1.000	1.000	1.000	0.020
σ_I 0.5%	0.10	0.930	1.000	1.000	0.002
σ_I 1%	0.10	0.932	1.000	1.000	0.002
σ_I 2%	0.08	0.908	1.000	1.000	0.001
Fibre 25 km	0.10	1.000	1.000	1.000	0.000
Fibre 50 km	0.10	0.932	1.000	1.000	0.002
Fibre 75 km	0.11	0.210	1.000	1.000	0.000

role to attackers who leave a statistical trace. Detection accuracy on intercept-resend or PNS, the headline numbers of several ML studies [15, 16], measures something the protocol already provides. Its operational value is real but different: faster incident response, and a signal that separates a denial-of-service from a degrading fibre. Reporting it as a security gain overstates it.

8.2 Why the learned detectors fail under drift

Drift is not noise around a fixed point. Misalignment grows for an hour and resets; loss follows the day. A model trained on stationary data learns a tight region around the calibration point, and the healthy link leaves that region within minutes. Domain randomisation widens the region but blurs the boundary to the attack classes, so RF-DR still raised false alarms at 9.16% at $d = 1$ with a seed-to-seed interval wider than the mean. IF-drift shows the other side: with representative benign drift it stays at 0.34% at $d = 1$, within a small factor of the target, and fails once the drift leaves the range it was fitted on. The DCS avoids the problem by design rather than by data. It refits the three drifting parameters in every window, and only relations that drift cannot change enter the statistic. The cost is visible too. Whatever the refit can absorb, the statistic cannot see: pure loss, and the missing dark counts of Blind-nodark. The post-hoc DCS+Y0 variant shows that adding prior knowledge about one parameter recovers an attack and pays for it in false alarms when that parameter drifts.

8.3 The limits of the indistinguishability argument

Proposition 1 is an existence statement about one attacker, and it assumes a photon-number-resolving detector for Eve, knowledge of the current channel and the ability to blind Bob's detectors without side effects. Each assumption is demanding in practice, and each has been met in some form: blinding of commercial detectors [4] and a full-field eavesdropper that left the QBER unchanged [5]. The static variant shows that partial knowledge already creates a trace, though in an unexpected direction: an attacker who replays a stationary channel is caught by the absence of drift, not by its presence. A defender can exploit this only with a detector that alarms on low variation, and a capable attacker removes the trace by tracking the channel.

The practical conclusion is not new, but the benchmark makes it quantitative: against detector-control attacks, telemetry is a weak line of defence. Hardware measures such as photocurrent monitoring [21], randomised detector efficiency [22], with the caveats found in testing [29], or measurement-device-independent protocols [20] change the observation model itself, and only they can turn Blind-cal into a detectable attack. A telemetry monitor should be judged by its false-alarm rate on a drifting link and by the model-violating attacks it catches, with an explicit statement of the attacker it cannot catch.

8.4 Threats to validity

Internal. All results come from one simulator. The device model (2) omits afterpulsing, detector dead time, efficiency mismatch and basis-dependent misalignment, so some benign effects that would raise DCS residuals in hardware are absent; the empirical-null standardisation is meant to absorb such effects, but its behaviour on real data is untested. Each attack is a model of the attack, not an implementation. The breach test (5) is one-sided and counts only bits Eve knows exactly.

External. The drift amplitudes are assumptions guided by qualitative reports [10, 11], not fitted to a link; the scale d and the sensitivity runs bracket them but cannot replace field data. The protocol is three-intensity decoy-state BB84 with one finite-key analysis [18]; other protocols and tighter analyses [24] would change key rates but not the indistinguishability argument, which depends only on the photon-number structure.

Construct. Per-window FPR at a fixed threshold is one operating point. Sequential procedures such as CUSUM [33] trade delay against false alarms differently and could raise power for small intercept-resend fractions; the comparison here is window-level for all detectors alike. The ML baselines follow the feature sets and training styles of the cited work but are not reimplementations of those papers.

Statistical. Ten seeds bound the attainable Wilcoxon p-values, so effect sizes are reported as intervals. The AUC near 0.5 for Blind-cal is an estimate with sampling spread (0.47–0.55), consistent with the exact equality that Proposition 1 proves.

9 Conclusion

Window telemetry cannot certify the absence of a detector-control attack. A faked-state attacker who reproduces the honest photon-number yields and error probabilities leaves the joint law of

all counts unchanged, so every telemetry detector, learned or physical, works at chance while the finite-key analysis outputs a key the attacker knows. Against attacks inside the device model, the decoy-state bound already protects the key, and detection serves operations rather than secrecy.

Between those two cases lies the useful work of a monitor: catching attackers whose calibration is imperfect without raising alarms on a healthy, drifting link. Here the benchmark separates the detectors sharply. Classifiers and isolation forests calibrated on stationary data alarmed in more than 97% of benign windows under nominal drift; domain randomisation reduced but did not remove the effect. A decoy-consistency statistic that refits the drifting parameters per window held the false-alarm rate near its 10^{-3} target up to fourfold drift and flagged naive and gain-only blinding in the first window, but it misses attacks that the refit absorbs. Future work should test the statistic on recorded telemetry from a field link, add sequential decision rules, and couple telemetry monitoring with hardware countermeasures that change what an attacker can reproduce. Code and per-seed results are released with this article, together with the manuscript source.

Data availability statement

The simulator, detectors, finite-key code, per-seed results and the scripts that produce every table and figure are openly available under the MIT licence at `https://github.com/osmancancet/qkd-telemetry-ids`.

A Supplementary tables

Table 7: Per-window false-alarm rate (%) when thresholds are calibrated on benign data with drift ($d \in \{0.5, 1, 2\}$). RF-stat never alarms because its benign scores under drift saturate at 1, which also removes all detection.

Detector	$d = 0$	$d = 1$	$d = 2$	$d = 4$
QBER>11%	0.00	0.00	0.00	0.00
DCS (ours)	0.09±0.02	0.10±0.03	0.11±0.02	0.18±0.05
IF-stat	0.00	0.04±0.03	0.29±0.10	3.58±1.42
IF-drift	0.00	0.00	0.54±0.32	22.17±3.59
RF-stat	0.00	0.00	0.00	0.00
RF-DR	0.00	0.00	0.24±0.23	5.88±6.08

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Table 8: ROC AUC at $d = 0$ (mean over 10 seeds).

Attack	QBER>11%	DCS (ours)	IF-stat	IF-drift	RF-stat	RF-DR
IR, $f = 0.05$	1.00	1.00	1.00	0.91	1.00	0.99
IR, $f = 0.2$	1.00	1.00	1.00	1.00	1.00	1.00
PNS	0.49	1.00	1.00	1.00	1.00	1.00
Blind-naive	0.00	1.00	1.00	1.00	1.00	1.00
Blind-gain	0.00	1.00	1.00	1.00	1.00	1.00
Blind-nodark	0.07	0.55	1.00	1.00	1.00	1.00
Blind-cal	0.49	0.50	0.50	0.50	0.50	0.50
Blind-cal-static	0.50	0.49	0.50	0.50	0.50	0.50
Loss +1 dB	0.63	0.48	1.00	1.00	1.00	1.00
Loss +3 dB	0.88	0.44	1.00	1.00	1.00	1.00

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